Project Title Smart Expiry Management System (Expiry Tracker)

1. Objective

The main goal of the Smart Expiry Management System (Expiry Tracker) is to build an automated system to help users to manage and monitor the expiry dates of products e ectively.

The goals of the project are:

•To design an online platform where product information like product name, product type, date of manufacture and date of expiry will be stored.

•To automatically track the expiry status of products.

•To give color-based alerts depending upon the number of days left till expiry of the product:

•Green Alert: More than 30 days left till expiry of the product.

•Yellow Alert: 30 days or less left till expiry of the product.

•Red Alert: The product has been expired.

•To remind users of products nearing their expiry date.

•To minimize the wastage of products and avoid the usage of expired products.

•To maintain inventory and keep the record of the products.

2. Scope of the Project

The Smart Expiry Management System can be applied both in domestic and commercial settings where there is a need to track items that have expiration dates.

Scope of the project is:

•Household Management: Management of expiry date of foods, medicines, and hygiene items in households.

•Medical Stores & Pharmacies: Management of medicines and ensuring no expired drugs are sold.

•Hospitals & Clinics: Management of medical items and equipment with expiry dates.

•Supermarkets & Retailers: Management of huge inventories and minimizing loss of products.

•Food Industries: Management of packaged food items and perishable items.

•Warehouse Management: Maintenance of well-maintained stock records.

The system aims to minimize manual e orts, enhance accuracy, and provide timely notifications for e icient management of products.

3. Methodology

The designing of the Smart Expiry Management System has been done in a systematic way and can be elaborated as follows:

1. Requirement analysis

Requirement analysis is performed in order to get user requirements related to product management, expiry management, and notifications.

2. System design

The design of the system is performed according to the requirement of the project.

3. Database design

Database is designed in order to store data related to users, products, expiry date, categories and notification.

4. Application Development

Front-end and back-end development is performed in order to allow user interaction and system functionality.

5. Expiry calculation and Notification generation

The system calculates the remaining days to expiry and assign notifications accordingly:

•Green: More than 30 days left.

•Yellow: 30 days or less left.

•Red: Expired products.

6. Testing

Testing of the system is performed in order to make sure that all functionalities work properly, such as login, product management, expiry management and notifications.

7. Deployment and maintenance

After successful testing of the system it can be deployed for practical use and maintained if needed.

4. Tools and Technologies

Hardware requirements

•Central Processor: Intel Core i3 or higher

•Memory: 4 GB minimum

•Hard Disk Space: 100 GB or higher

•Internet Connection

Software requirements

Front-end Technologies

•HTML5: Used to develop structure of web pages.

•CSS3: Used to design and enhance appearance of the application.

•JavaScript: Used to implement interactivity and client-side validations.

•Bootstrap: Used to develop responsive user interface.

Back-end Technology

•PHP: Used to develop server-side code to handle user request, manage products, expiry, and system logic.

Database

•MySQL: Used to manage user details, product details, expiry date, and expiry notification records.

Development tools

•Visual Studio Code: Used to develop the application.

•XAMPP: Used as a local server environment consisting of Apache, PHP, and MySQL.

Browsers Supported

•Google Chrome

5. Expected Outcomes

The expected outcomes of the Smart Expiry Management System are:

•A user-friendly platform for managing product expiry information.

•Automatic identification of products nearing expiry.

•Reduction in product wastage and financial losses.

•Prevention of usage and sale of expired products.

•Improved inventory organization and management.

•Timely reminders through alerts and notifications.

•Easy monitoring of product status through dashboards and reports.

6.Conclusion

The **Smart Expiry Management System** is a practical and user-friendly solution that helps users monitor the expiry dates of medicines, groceries, cosmetics, and other products. By providing timely reminders and maintaining product records, the system reduces wastage, saves money, and promotes health and safety. It also minimizes the effort of manually checking expiry dates. With future enhancements such as barcode scanning, cloud synchronization, and AI-based expiry predictions, the system can become an even more efficient and reliable tool for homes, medical stores, supermarkets, and small businesses.